GEK-83854

DRIVE SYSTEM INSTRUCTIONS

THERMAL GAS ANALYZER EQUIPMENT

228B8352

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL@ ELECTRIC

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INTRODUCTION

These instructions cover the installation, operation and adjustment of the Thermal Gas Analyzer Equipment. This equipment is designed to monitor and indicate hydrogen purity.

The equipment consists of several parts: a power supply panel, a detector panel, a cell block, a purity indicator, a filter-dryer unit, a milliammeter and a balanced-bridge current adjusting rheostat. The purity indicator gives a continuous reading of hydrogen purity by indicating percent hydrogen in air over a range of 50% to 100%. It includes high and low alarm set-pointers, adjustable to any point on the scale. The purity indicator is calibrated for use when a second or remote purity indicator is used. The second or remote purity indicator is generally located in the control room and is not equipped with alarm contacts. The filter-dryer unit is used to eliminate foreign particales and moisture from the gas. The milliammeters and bridge current rheostats are used for calibration. The equipment operates on a 120 volt, 50/60 Hz single phase power source. The thermal gas analyzer should be mounted in a well ventilated area.

RECEIVING, HANDLING AND STORAGE

Receiving and Handling

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts from being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediatley with the transportation company and the nearest General Electric Sales Offices should be notified promptly.

Storage

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for an extended period of time, it should be kept heated to at least 5°C above ambient temperature to keep out moisture.

DESCRIPTION

General

The thermal conductivity of a gas is a measure of its ability to conduct heat and is defined as the "time rate of transfer of heat by conduction through a unit thickness of gas across a unit area for a unit difference in temperature". In the metric system, it is measured in calories per second per square centimeter for a difference in temperature of 1°C. In thermal conductivity work, the relative conductivity, or the ratio of conductivity of a gas to the conductivity of air, is often used. In relative conductivity, two gasses: helium and hydrogen, stand apart, having conduction about 6 to 7 times as great as air.

The high thermal conductivity of hydrogen makes it possible to run large hydrogen-cooled generators at higher loads for a given temperature rise because more rapid removal of heat from the winding and iron structure is possible.

High thermal conductivity is also an excellent indicator of hydrogen purity because a mixture of any gas with hydrogen will register lower thermal conductivity than hydrogen gas. About 10 percent helium in hydrogen would have about the same thermal conductivity as one percent air in hydrogen. (Helium has about 90 percent of the conductivity of hydrogen.) This, however, is not a power station problem, since there are no ordinary sources of helium to contaminate hydrogen in the usual system. All other gases have about the same thermal conductivity as air, although air is the only contaminant usually present. Analyzers are calibrated for percent of hydrogen in air.

Electrical Equipment

Figure 1 shows the essential parts of the thermal gas analyzer equipment. The power supply panel is supplied with 120 volts. A gas sample is introduced into the cell block which is electrically connected to the detector panel. Bridge current is adjusted by the bridge-current rheostat and indicated by the bridge-current milliammeter. The percent hydrogen in air is indicated by the type 195 (meter-relay) purity indicator. Adjustable high and low alarm set-pointers on the indicator unit initiate the control unit relays of the type 195 (meter-relay) purity indicator at predetermined set points.

Piping Circuit

Figure 2 shows the piping circuit for the thermal gas analyzer equipment. The cell block receives gas from the generator. The gas flow is regulated by the throttling valve associated with the flowmeter. Flow should be regulated as explained under Flow Control Section.

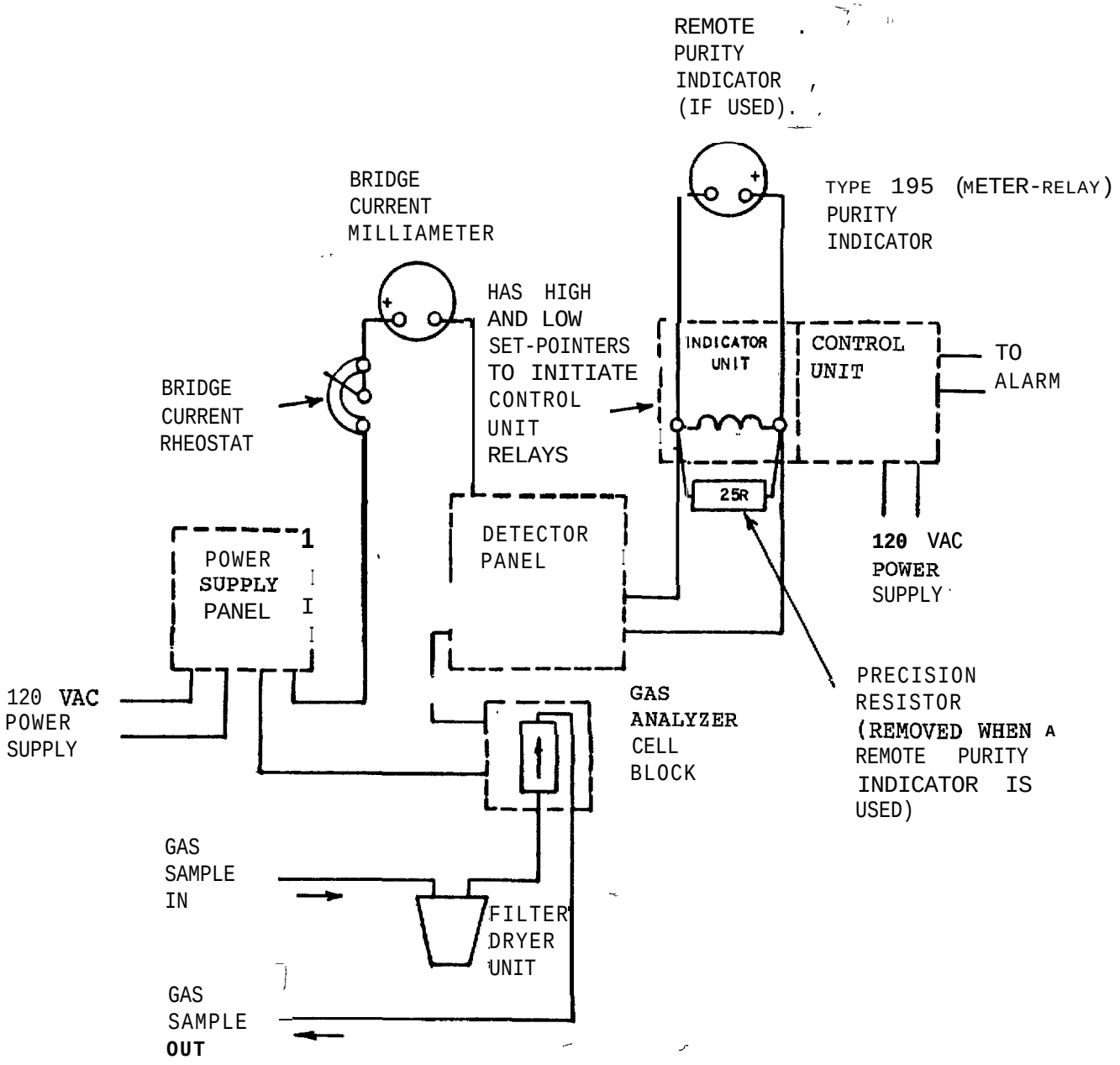


FIGURE 1
Simplified Block Diagram of Electrical Equipment

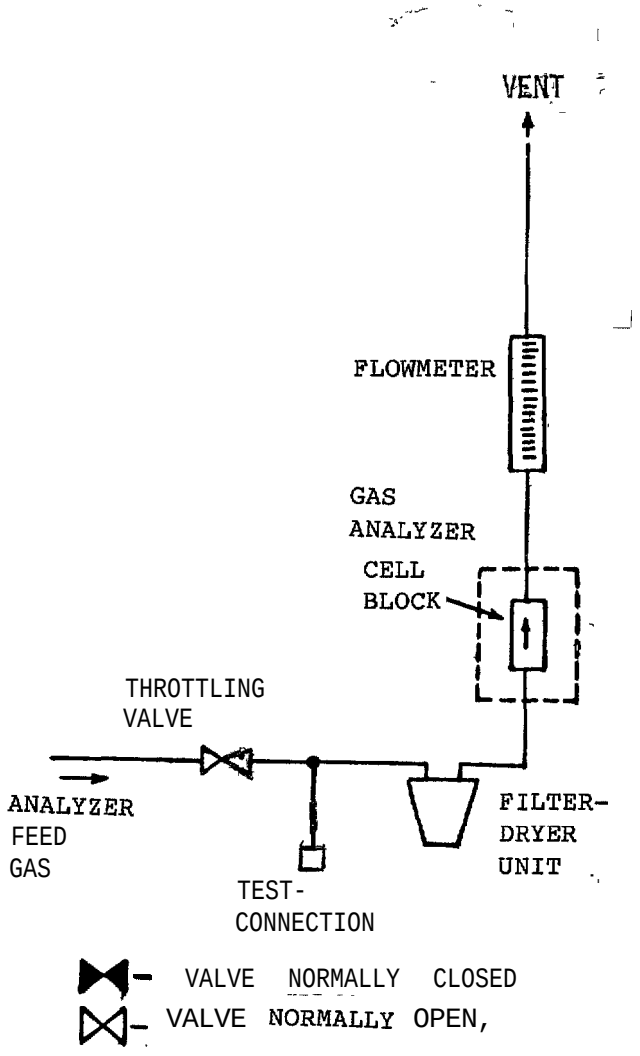


Figure 2
Piping Circuit

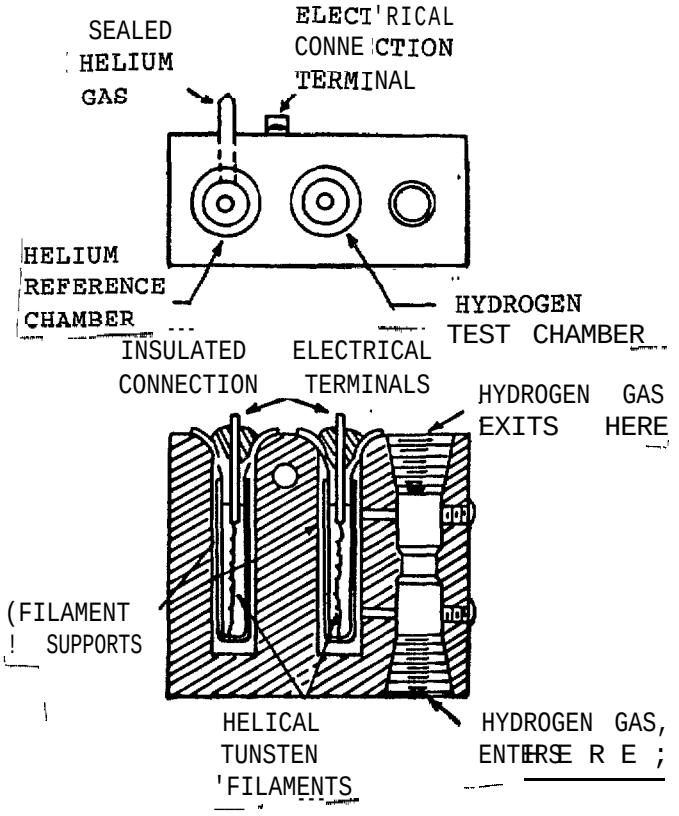


Figure 3
Typical Cell Block Construction

Gas Analysis Filaments

Figure 3 illustrates the construction of a typical cell block. A fine helical wire filament is placed axially in each of two cylindrical chambers in the metal cell block. One end of each filament is mechanically supported by the filament support structure and is thereby electrically connected to the main section of the cell block. Provision is made in the cell block for the hydrogen gas to be admitted to the hydrogen test chamber. Both filaments are heated by an electric current. Some of the heat is lost by conduction to the filament supports and by radiation and convection, but most of the heat is transferred from the filament to the cell structure by conduction through the gas surrounding the filament. The temperature and therefore the resistance of each filament is then an inverse function of the thermal conductivity of the gas surrounding the filament. The thermal conductivity of the hydrogen increases with its purity. The temperature and resistance of the filament in the hydrogen chamber will therefore decrease as the hydrogen purity of the gas surroundig the filament increases. Therefore, if the filament is used as one leg in a Wheatstone bridge, the change in resistance of the filament as the gas purity changes can be used to indicate the hydrogen purity of a sample of gas passed through the hydrogen test cell chamber containing the hydrogen filament. The helium reference filament maintains essentially constant resistance since the filament is surrounded by a sealed helium gas. Variations in ambient temperature and other heating effects are approximatley cancelled since both filaments are mounted in the same metal enclosure.

Electrical Circuit

Figure 4 illustrates the electrical circuit of the thermal gas analyzer equipment. A 120 volt AC supply is connected through the on-off switch 1SW and fuses 1FU and 2FU to voltage transformer 1T. The output of 1T is rectified by a full-wave silicon rectifier bridge 1 REC. The output voltage of this rectifier bridge is dropped across resistor 1R and regulated by zener 1Z. The regulated voltage is the DC supply for the Wheatstone bridge circuit. The essential components of the bridge circuit are an H2 analyzer filament, an He reference filament, bridge resistors 2R, 3R, 4R, and bridge balance rheostat 2P.

If the potentiometer 2P is adjusted so that the series-parallel combination of resistors 3R and 4R is equal to the resistance of 2R, the bridge will be in balance if the resistance of the H2 analyzer filament is equal to the resistance of the He reference filament.

This means that no voltage potential appears across the output terminals of the bridge AB. If the H2 analyzer filament resistance decreases, a positive voltage will appear at terminal B. A current will then flow from terminal (B) through the purity indicator 63-4X and the bridge output rheostat 3P to terminal (A) of the bridge circuit. The more the decrease of the H_2 analyzer filament resistance, the more current flows through the purity indicator. The bridge circuit is normally balanced at the factory so that no current flows through the purity indicator when a mixture of 50% hydrogen and 50% nitrogen gas is passing through the analyzer cell. With this condition, the output of the bridge is zero and the purity indicator will read 50% hydrogn in air. This corresponds to zero current through the purity indicator. The bridge balance potentimeter 2P is used to make a fine adjustment of this point. As the purity of the gas increases toward 100% hydrogen, the resistance of the hydrogen analyzer filament will decrease and current will flow through the purity indicator from plus to minus. When the test gas is 100% hydrogen, the purity indicator will read 100% hydrogen in air. This corresponds to 1 milliampere through the purity indicator. The bridge output potentiometer 3P is used to make a fine adjustment of this point. The bridge current flowing through milliameter M-l is 325 milliamperes. The exact value of the bridge current is not critical but the current must not change after the gas analyzer is calibrated. The effects of an increase in this current are to increase the sensitivity of the purity indicator to changes in hydrogen purity and cause the scale readings to expand outward from the scale. Readings on the lower end of the region of 60 to 80 percent hydrogen in air will be effected only slightly. A decrease in bridge current will cause the upper end of the scale to read low and the lower end of the scale to read high.

Remote Purity Indicators

All purity indicators are calibrated for use with a remote purity indicator. If a remote purity indicator is required for indication of the generator-hydrogen purity at a location remote from the gas analyzer equipment, remove resistor 25R connected in parallel with the (type 195 meter-relay) purity indicator.

Filter-Dryer Unit (3S7576SA107)

The filter-dryer unit uses a replaceable desiccant purifier cartridge shipped in a hermetically sealed can. This cartridge must be replaced if the drying agent becomes saturated by moisture.

The moisture and impurity indicator indicates the presence of moisture in the gas line and the subsequent exhaustion of the desiccant in the purifier. The indicating element is filled with silica gel. The normal blue color of this desiccant agent turns to pink when expended. The expended desiccant agent may be regenerated by heating the indicating element in an oven at 150°C for 2 or 3 hours. A manufacturers knurled-cap plug assembly which contains the desiccant is available as a replacement for this unit.

It is usually possible to keep the generator hydrogen at a low relative humidity so that frequent replacement of the desiccant materials is not necessary.

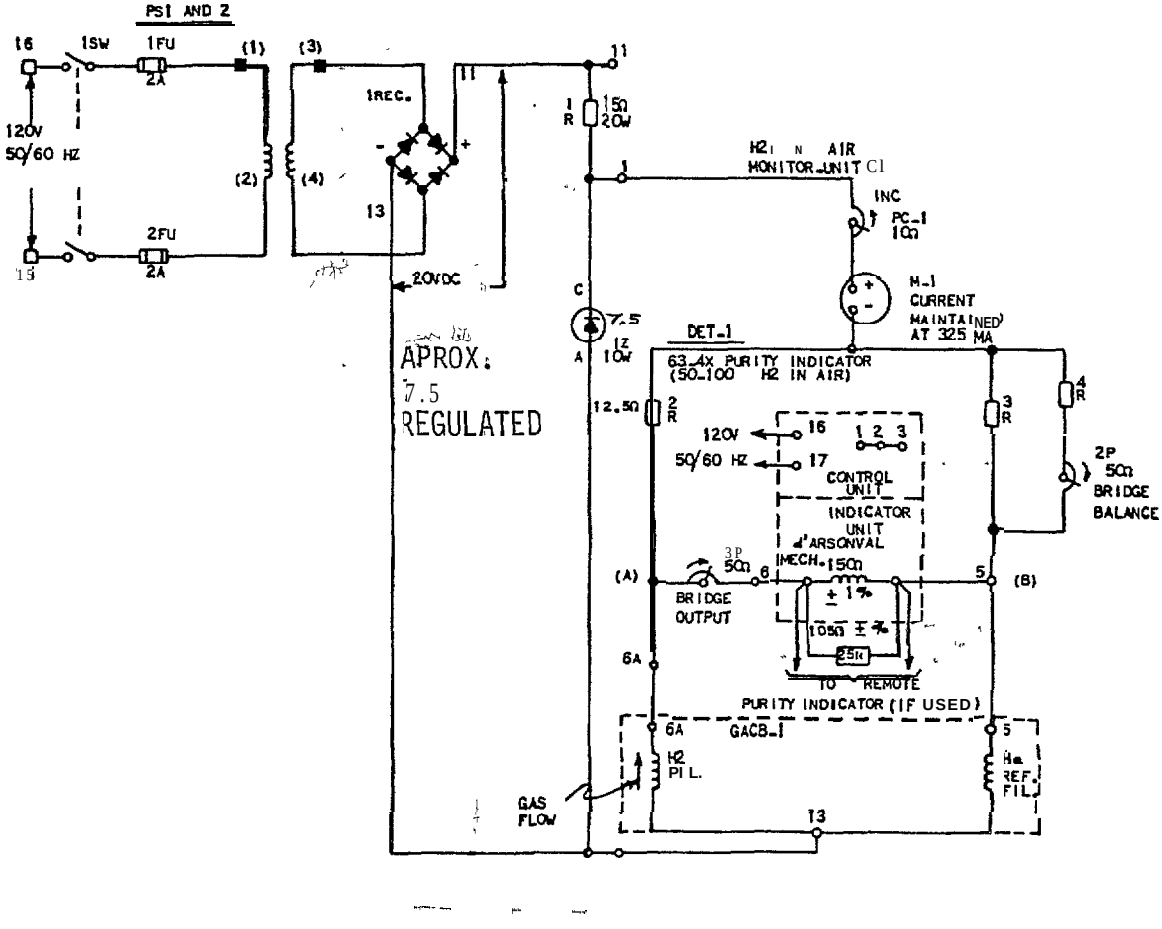


Figure 4
Electrical Circuit

INSTALLATION

Location and Mounting

The cabinet should be mounted in a clean, well ventilated vibration free area. The cabinet should be mounted in an upright position. See figure 6 for cabinet outline.

WARNING

PURE HYDROGEN IS NORMALLY NONFLAMMABLE. HOWEVER, A MIXTURE BETWEEN 5 AND 75 PERCENT HYDROGEN IN AIR WILL EXPLODE IF IGNITED IN AN ENCLOSED SPACE. THE GAS ANALYZER EQUIPMENT CAN BE SAFELY OPERATED BY MAINTAINING A HYDROGEN PURITY IN THE DETECTOR BLOCK BETWEEN 97 AND 98 PERCENT HYDROGEN IN AIR AT ALL TIMES, THUS MAKING AN EXPLOSION NORMALLY IMPOSSIBLE, AND BY MOUNTING THE GAS ANALYZER EQUIPMENT IN A PROPERLY VENTIALTED CONTROL CABINET IN WHICH ALL ELECTRICAL CONTROLS WHICH MIGHT CAUSE AN ARC ARE EITHER BEHIND A GAS TIGHT SEAL OR ARE HERMETICALLY SEALED-TYPE SWITCHES. THIS LATTER PRECAUTION GREATLY REDUCES THE POSSIBILITY OF A FIRE FROM ANY LEAK IN THE GAS PIPING WHICH COULD COLLECT AND BUILD UP TO AN EXPLOSIVE MIXTURE.

Connections

Connections must be made in accordance with the diagram supplied with the equipment. Care must be exercised to determine that connections are correct to avoid damaging the equipment.

OPERATION AND ADJUSTMENTS

The following steps are required to place the equipment in operation:

1. Regulate the flow as described under the Flow Control section.
2. Adjust potentiometer PC-l so the bridge-current is at the red line mark (325 milliamperes) on the bridge-current milliammeter M-l.
3. Readings of the purity of the gas may now be taken from the type 195 (meter-relay) purity indicator.
4. The high and low alarm set-pointers on the type 195 (meter-relay) purity indicator may be set to any point by adjusting the two knobs located on the front of the instrument.

The gas analyzer detector panel will require no attention during operation except to check flow rate and bridge-current immediatley before taking readings. Large changes in gas concentration will cause minor changes in bridge-current.

Thermal Gas Analyzer Calibration

General

It sometimes becomes necessary, due to the replacement of a component such as a purity indicator or a cell block, etc. to recalibrate the thermal gas analyzer equipment. This can be accomplished by either of two methods. The method used depends on the component being replaced and is described below.

Approximate Recalibration Method

This first method is used after a minor fault has been corrected: An example of a minor fault is a sticking needle on the purity indicator.

Recalibration is accomplished by purging the test cell block with a commercially pure hydrogen gas and adjusting potentiometer 3P until the purity indicator reads 99.6%. This method of calibration eliminates any error at the upper end of the scale.

Any error existing at the 50 percent point will be reduced proportionally as scale readings increase, becoming zero at the upper end of the scale. For example, an error at the 50 percent point would be reduced to one-tenth of the error when the pointer reads 95 percent hydrogen in air.

The purity indicator is accurate only when the bridge current is set to the red line of the milliammeter scale (325 milliamperes). This is particularly true at low-scale deflection. At the center and high end of the scale, readings will appear independent of bridge-current.

The speed at which the purity indicator reads the purity of the gas is largely a matter of the capacity of the line feeding the instrument. It is permissible to run rapidly through the instruments if the readings are disregarded during the high-flow. When the flow is reduced to normal, the purity indicator will read the correct value of the gas within a few seconds.

Water vapor will cause some changes in the reading. Machine gas is usually dry enough so that effects of water vapor are negligible. Free water in a cell should be avoided, as it will change the calibration, and water contamination with oil and other substances may permanently damage the cell.

Exact Recalibration Method

The second method of calibration is used after a major fault has been corrected. An example of a major fault is the failure of the cell block.

The following procedure is recommended for recalibration:

Figure 3 shows a typical cell block construction. A study of this construction will pinpoint the exact locations of the H2 and He helical tungsten filaments. These filaments are shown in the electrical circuit (Figure 4).

It is suggested that a tank of 50 to 55% hydrogen-nitrogen gas be available.

The calibration of the gas analyzer equipment is accomplished by balancing the Wheatstone bridge, points (A) and (B), to zero output with the 50% hydrogen, and maximum output (full scale) with 100% hydrogen in the analyzer cell.

In practice, it is usually not possible to obtain an exact 50% hydrogen-nitrogen mixture. In this event, all zero output adjustments should be made at the percent hydrogen labeled on the tank. That is, for a 52% gydrogen-nitrogen gas mixture, all zero output adjustments should be made at the 52% mark on the purity indicator scale.

To simplify the calibration procedures, all adjustments will be made assuming use of a 50% hydrogen-nitrogen mixture. All circuit references will be made using the electrical circuit (Figure 4).

Testing Sequence

1. Install the cell block in place of the defective unit.

2. Adjust the sliders of potentiometers 2P and 3P, located in the detector panel, in the mid-position.

3. Regulate the flow of the 50% hydrogen-nitrogen gas, in the cell block, as described under Flow Control.

4. When the cell block is charged with the 50% hydrogen mixture, adjust PC-1 until the milliammeter in the bridge circuit reads 325 milliamperes (red line). Then, adjust potentiometer 2P until the needle on the purity indicator reads at the 50% mark. With the calibration complete for the 50% point on the purity indicator, purge the cell block with 100% hydrogen gas.

5. When the cell block is charged with the 100% hydrogen gas, adjust the milliammeter in the bridge circuit to read 325 milliamperes (red line). The adjustment of the current should always be made when switching from 50 to 100% hydrogen gas or vice versa.

6. Adjust potentiometer 3P (clockwise), locatd in the detector panel, until the needle of the purity indicator reads at the 100% mark.

7. To assure accuracy of adjustment, repeat the above steps using the 50 and 100% hydrogen gas until there are two repetitive readings.

Flow Control

A separate Flowrator* is used as a flow indicator for the gas analyzer cell block. (See Figure 2). Normally, the hydrogen flow is regulated by adjusting the throttling valve until approximately 1.0 STD. cubic foot per hour or 470 cubic centimeters per minute, at 0.5 PSIG, at 70°F of pure hydrogen gas, is maintained on the Flowrator.

Flow-Error

Flow should always be adjusted below the level at which flow-error becomes apparent. Flow error can be determined by the purity indicator readings. If the flow-rate is above normal, the purity indicator will tend to read on the high side. For example: With a 50% hydrogen gas in the system, and an abnormal flow-rate, the purity indicator readings will read approximately 52 to 57% instead of 50% H_2-in-air.

Sensitivity to flow-error also depends on the conductivity of the gas, the error being less as the purity of the hydrogen increases. Individual cells may vary slightly in their sensitivity to flow. In any case, the maximum permissible flow may be determined by increasing flow until flow-error is evident. Sufficient time should be allowed to scavenge the pipe lines with the gas to be measured before readings are taken. High flow-rate may be used to scavenge the lines, but readings should not be taken until the flow is adjusted below the limits of flow-error.

Remote Purity Indicator

The remote purity indicator must be adjusted to full scale while it is in parallel with the type 195 (meter-relay) purity indicator. One milliampere at 105 millivolts from the thermal gas analyzer bridge is used to adjust the remote purity indicator with the type 195 (meter-relay) purity indicator. This may be obtained from the gas analyzer by filling the cell with pure hydrogen gas and adjusting the bridge-current to 325 milliamperes.

Maintain the reading of the type 195 (meter-relay) purity indicator at 100% by adjusting the bridge-current output potentiometer 3P, and adjust the remote purity indicator to read 100% hydrogen in air by adjusting the rheostat loacted in the remote purity indicator.

*Trademark of Fischer and Porter Company, Warminster, Pennsylvania

MAINTENENACE

Routine maintenance is limited to changing the expended activated alumina in the filter dryer units.

TROUBLESHOOTING

General

In the event of abnormal operation of any part of the Thermal Gas Analyzer equipment, the equipment should be immediately removed from service and carefully tested to determine the cause of difficulty. The instructions previously given should be referred to, and if any changes in connections are made, the circuits affected should be fully tested, in accordance with the applicable section of these instructions, before returning the equipment to service.

Troubleshooting Chart

TROUBLE	PROBABLE CAUSE AND REMEDY
Purity Indicator moves off-scale to the right	Shorting or cooling of the cell filament by water, oil or other foreign matter. Check for source of trouble; when cleared, replace cell block; Sealed helium reference gas has leaked out. Replace cell block.
Bridge-current tends to decrease and may reach the point where it is no longer possible to set current to red line (325 milliamperes) on milliammeter.	Sealed helium reference gas has leaked out. Replace cell block.
Bridge current increases	Shorted cell block filaments. Replace cell block.
Irregular movement of the needle in the purity indicator when the balance-bridge potentiometer 2P is adjusted.	Broken wires on 2P. Replace 2P. Irregular contact resistance on 2P caused by dust and lint. Replace 2P.
Discontinuity of jump of needle in purity indicator.	Broken wires on 2P. Replace 2P
Purity indicator drops to its mechanical zero.	Broken wires on 2P or 3P (when slider contact crosses over broken wire). Replace 2P or 3P.
Purity indicator readings become unsteady and then shift, indicating an apparent increase in hydrogen purity.	Flow has increased beyond a certain amount. Shut off flow for short time, gas analyzer will immediately assume correct reading.

RENEWAL PARTS

When ordering renewal parts, address the nearest Sales Office of General Electric Company, specify the quantity required, and describe the required parts in detail. In addition, give the 3s number and complete the nameplate rating of the equipment.

PRINCIPAL RENEWAL PARTS LIST FOR: 22888352

DIAGRAM SYMBOL	QUANTITY	ORDERING NUMBER	DESCRIPTION
PS1 & 2	1	3S7576SA200C1	Power Panel
1T	1	44B334523-G01	Transformer
1REC	1	44B330140-G03	Comp. Board Assembly
	4	44B232019-005	Silicon Rectifier
	2	44A212119-001	Fuse Holder
1FU, 2FU	2	LFC#312002	Fuse, 2 Amp.
1R	1	A20C15CA/X101AC	Resistor; 15 Ohm, 2ow
1Z	1	68A7233P02BN	Zener, 7.5V, 10W
1SW	1	138A1031P4	Toggle Switch
DET-1	1	3S7576SA202C1	Detector Panel
2P, 3P	2	44A334273-001	Pot., 50 Ohm, 3.5W
2R	1	124A1892D125	Res., 12.5 Ohm
3R	1	124A1892D15	Res., 15 Ohm
4R	1	124A1892D30	Res., 30 Ohm
63-4X	1	44B218530-001	Purity Indicator
25R	1	983B585E105	Res., 105 Ohm, 1 W
M-1	1	933B579P2	Milliammeter, (Red Line at 325 Milliamperes)
PC-1	1	44A370848-D10	Pot., 10 Ohm, 2 W
		561A622P1	Knob for Potentiometer
GACB-1	1	3S7576SA103A8	Cell Block Assembly
	1	983B870G1 (1)	Cell Block
Flowmeter	1	205V210	Flowmeter
Valve	1	WCO#B-1KS4	Valve

PRINCIPAL RENEWAL PARTS LIST FOR: 228B8352

DIAGRAM SYMBOL	QUANTITY	ORDERING NUMBER	DESCRIPTION
	1	3S7576SA107A6	Filter-dryer unit
	1	44A333430-001	Purifier
	1	44A333431-001	Purifier Cartridge
	1	44A333432-001	Moisture & Impurity Indicator
	1	44A336755-001	For Replacement Knurled-Cap Plug Assembly with Desiccant

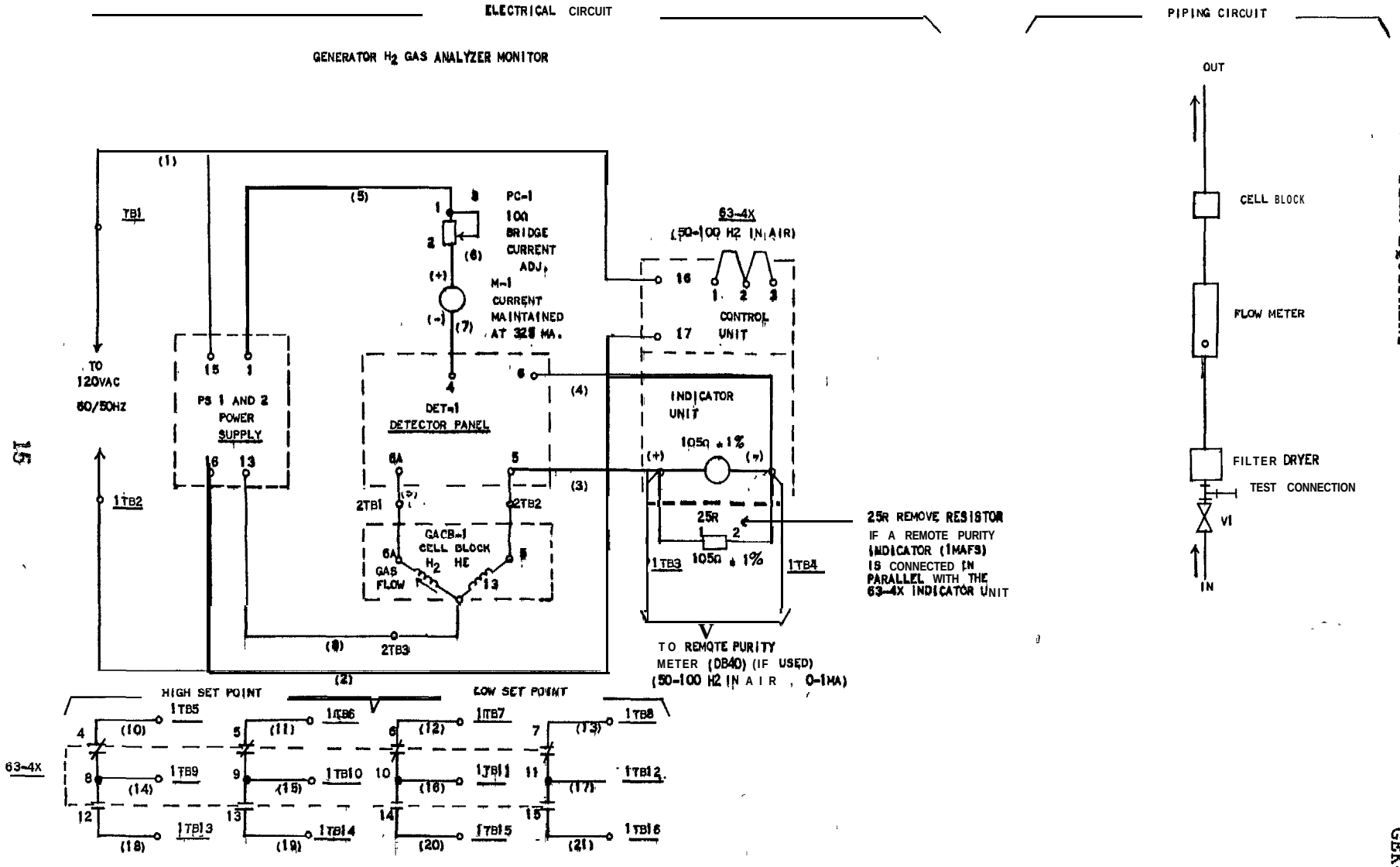
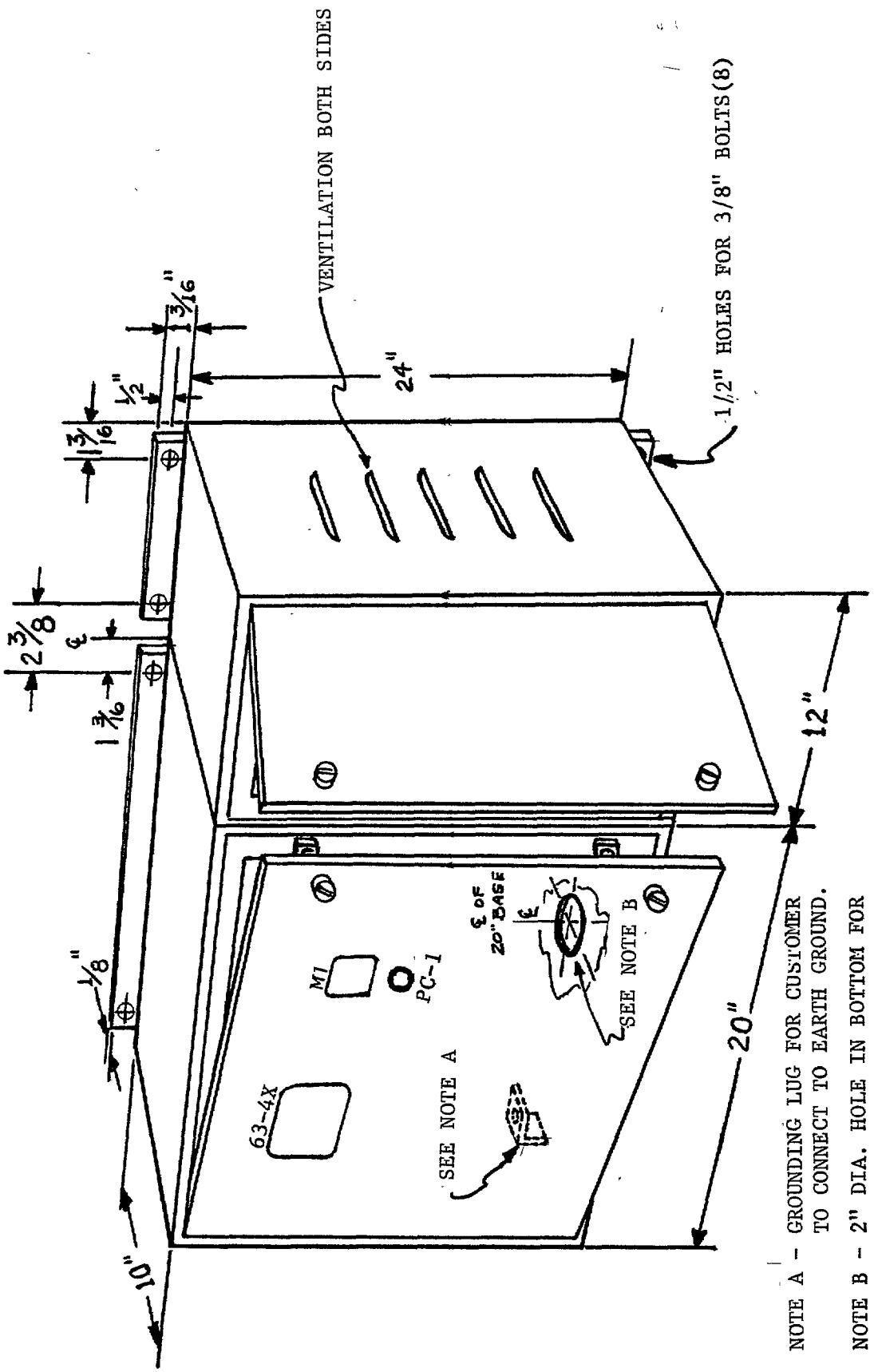


Figure 5 - Elementary Diagram



NOTE A - GROUNDING LUG FOR CUSTOMER TO CONNECT TO EARTH GROUND.

NOTE B - 2" DIA. HOLE IN BOTTOM FOR CUSTOMER CONDUIT.

FIGURE 6
OUTLINE

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